DATA STRUCTURES

NAME:
DATE: / /

Unit 1:- Introductions to Data structures

structure defination, initialisation, array as a structure, array within structure, union, understanding pointers, declaring and initialising pointers, accessing a variable and its pointer, static and dynamic memory allocation. defination of data structure, classification of data structures, primitive & non primitive operations on data structure, review of array.

Unit 2:- Searching and Sorting

Searching defination, searching techniques, sequential (linear) search, binary search. comparision between sequential and binary searching. Sorting defination sorting techniques - bubble sort, merge sort, selection sort, quick sort, Insertion sort.

Unit 3: Stacks and Queues

defination of stack array representation of stack, linked list, representation of stacks, operation performed on stacks, Infix, prefix, postfix notations, conversion of arithmetic expressions, application of stack. defination of Queue, array representation of Queue, types of Queues - simple Queue, circular queue, double ended queue, priority queue, operations on all the type of Queues.

Unit 4: linked list

Defination, representation of linked list in memory, types of linked list :- singly linked list, doubly linked list, Circular linked list, operations on linked list, creation, ~~insearch~~ insertion, deletion, search, display and traversing, advantages and disadvantages of linked list